Jia-Fong Yeh (葉佳峯)

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PhD in Computer Science at NTU with solid research experience in Machine Learning, Robotics, and Evolutionary Computation. My work addresses core challenges in real-world robotic applications, and has led to publications at top-tier venues such as ICLR, NeurIPS, and AAAI. I am driven to collaborate with international researchers, as demonstrated by my internship at Sony Japan. My research excellence and practical capabilities have been recognized through multiple national and academic awards.

Education

National Taiwan University (NTU) — *Taipei, Taiwan*

PH.D. IN COMPUTER SCIENCE AND INFORMATION ENGINEERING — *Sept. 2019 – July 2025*

- GPA: **4.21/4.3**
- Advisor: Prof. Winston H. Hsu

National Taiwan Normal University (NTNU) — *Taipei, Taiwan*

B.S. & M.S. IN COMPUTER SCIENCE AND INFORMATION ENGINEERING — *Sept. 2013 – June 2019*

- GPA: 3.74/4.3 & **4.3/4.3**
- Advisor: Prof. Tsung-Che Chiang and Prof. Shun-Shii Lin

Selected Research Projects

My research focuses on advancing **policy learning**, addressing **three major challenges**: (1) the complexity of the environment, (2) the lack of reward signals, and (3) the safety concerns on policy's behavior, as described below:

Learning Multi-stage Manipulation Tasks from Few Demonstrations | AAAI 2022

- Introduced an **attention-based** policy that learns from demonstrations on complex tasks
- Outperformed all baselines in **6 out of 8 cases**, with interpretable visualization results

Vision-Instruction Correlation Reward Generation | ICLR 2025

- Developed a hierarchical reward model using both **LLM** and **VLM** techniques
- Demonstrated a **43%** improvement over the strongest baseline in **long-horizon tasks**

Monitoring Policy's Behaviors in Unseen Environments | NeurIPS 2024

- Proposed novel **contrastive losses** to enhance monitoring for detecting erroneous actions
- Achieved the best results in **17 out of 21 cases**, with extensive ablations and visualizations

Experiences & Services

ML Research Intern, Sony Group Corporation — *Tokyo, Japan*

- Mentor: Dr. Wang Zhao and Mr. Shingo Takamatsu — *Oct. 2023 – Mar. 2024*
- Topic: *Reinforcement Learning with LLM/VLM guidance*

Area Chair

- ICLR'26

Reviewers

- CVPR, NeurIPS, ICML, ICLR, ICCV, AAAI, WACV, ICASSP, ICDL, ICMLW, CVPRW
- *Outstanding Reviewers*: CVPR'25

Skills

- **Programming Languages:** C/C++, Python, C#
- **ML Infrastructure/Packages:** PyTorch, TensorFlow, Weights&Biases, scikit-learn, AWS EC2
- **Robot Simulation:** Pybullet, Isaac Sim, Coppeliasim
- **Others** – Git, HTML, CSS, mysql, OpenMP
- **Selected Projects** –WebGL renderer [demo], Dark Chess AI, JPEG Decoder, Snake Game AI

Selected Awards

2025 **IPPR Dissertation Competition** (Honorable mention) [link]

2025 **TSC Thesis Award: AI Application Competition** [link]

2025 **Hon Hai Technology Award** [link]

2025 **NTUEE-1975 Technology Research Innovation Award** [link]

2025 **Phi Tau Phi Honorary Member** [link]

2024 **NeurIPS Scholar Award**

2022 **NOVATEK PhD Scholarship** [link]

2019 **ORSTW Master Thesis Award** (Honorable mention) [link]

2018 **ITSA Annual Collegiate Programming Contest** (Honorable mention) [link]

2016-2022 **Computer Chess Competition** (16 medals)

First-author Projects | citations: 420+ | h-index: 8 | i-10 index: 6

- VICtoR: Learning Hierarchical Vision-Instruction Correlation Rewards for Long-horizon Manipulation
  ICLR 2025 | co-first work | AR: 32.08% | **[PDF]**
- AED: Adaptable Error Detection for Few-shot Imitation Policy
  NeurIPS 2024 | AR: 25.8% | **[PDF]**
- Shared-unique Features and Task-aware Prioritized Sampling on Multi-task Reinforcement Learning
  arXiv 2024 | co-first work | **[PDF]**
- Stage Conscious Attention Network (SCAN): A Demonstration-conditioned Policy for Few-shot Imitation
  AAAI 2022 | co-first work | AR: 15% | **[PDF]**
- Large Margin Mechanism and Pseudo Query Set on Cross-Domain Few-Shot Learning
  CVPRW Report 2020 | citation: 20+ | **[PDF]**

- Modified L-SHADE for Single Objective Real-Parameter Optimization
  CEC 2019 | citation: 20+ | **[PDF]**
- Snake Game AI: Movement Rating Functions and Evolutionary Algorithm-based Optimization
  TAAI 2016 | **[PDF]**

I also have publications in *IEEE TMM*, *IEEE TAI*, *ICCV, EMNLP, CoRL*, *ICRA*, *ICASSP*, *BMVC*, *ICIP*, *ECCVW*, and *NeurIPSW*, please find them from my **Google Scholar**.